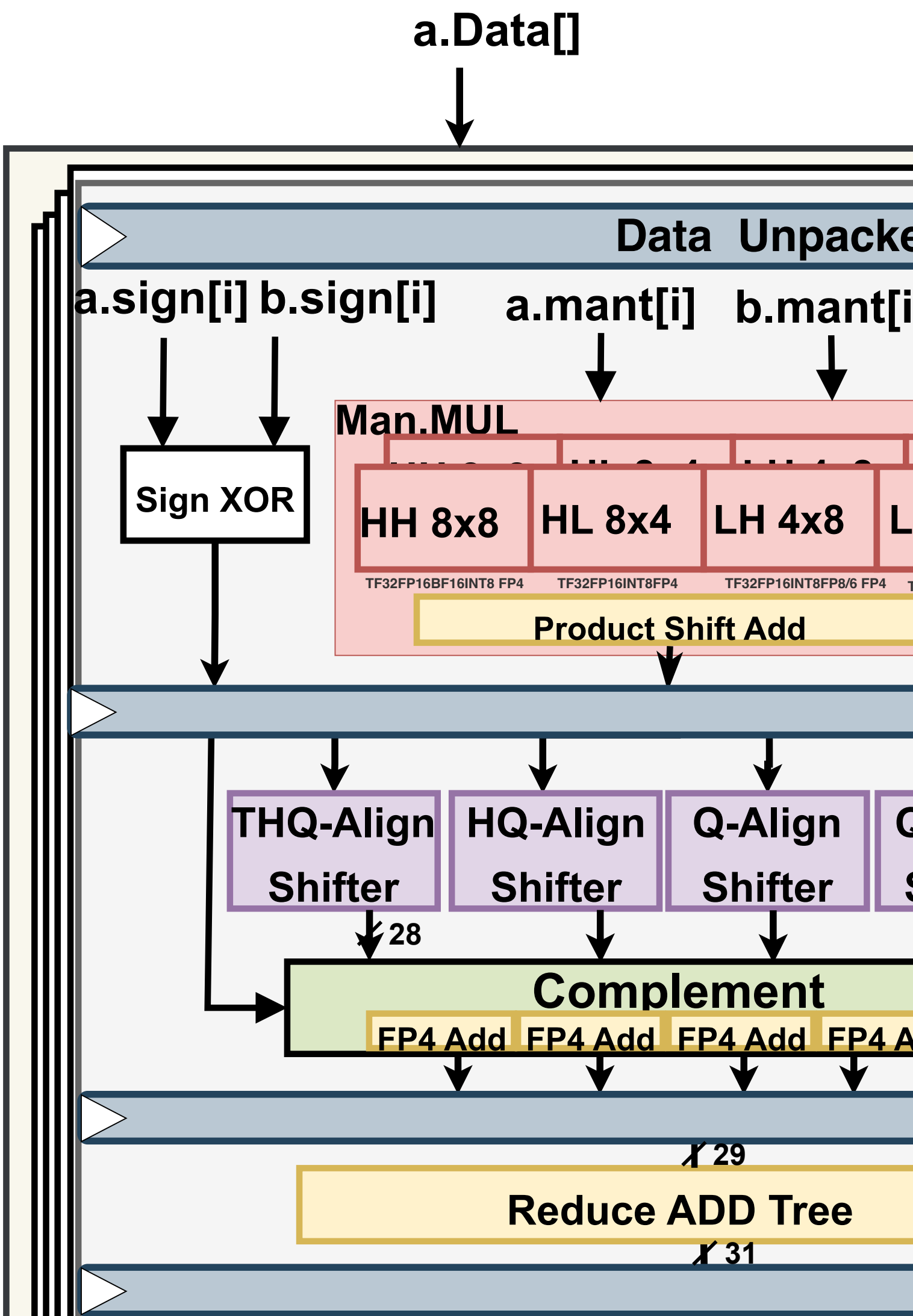
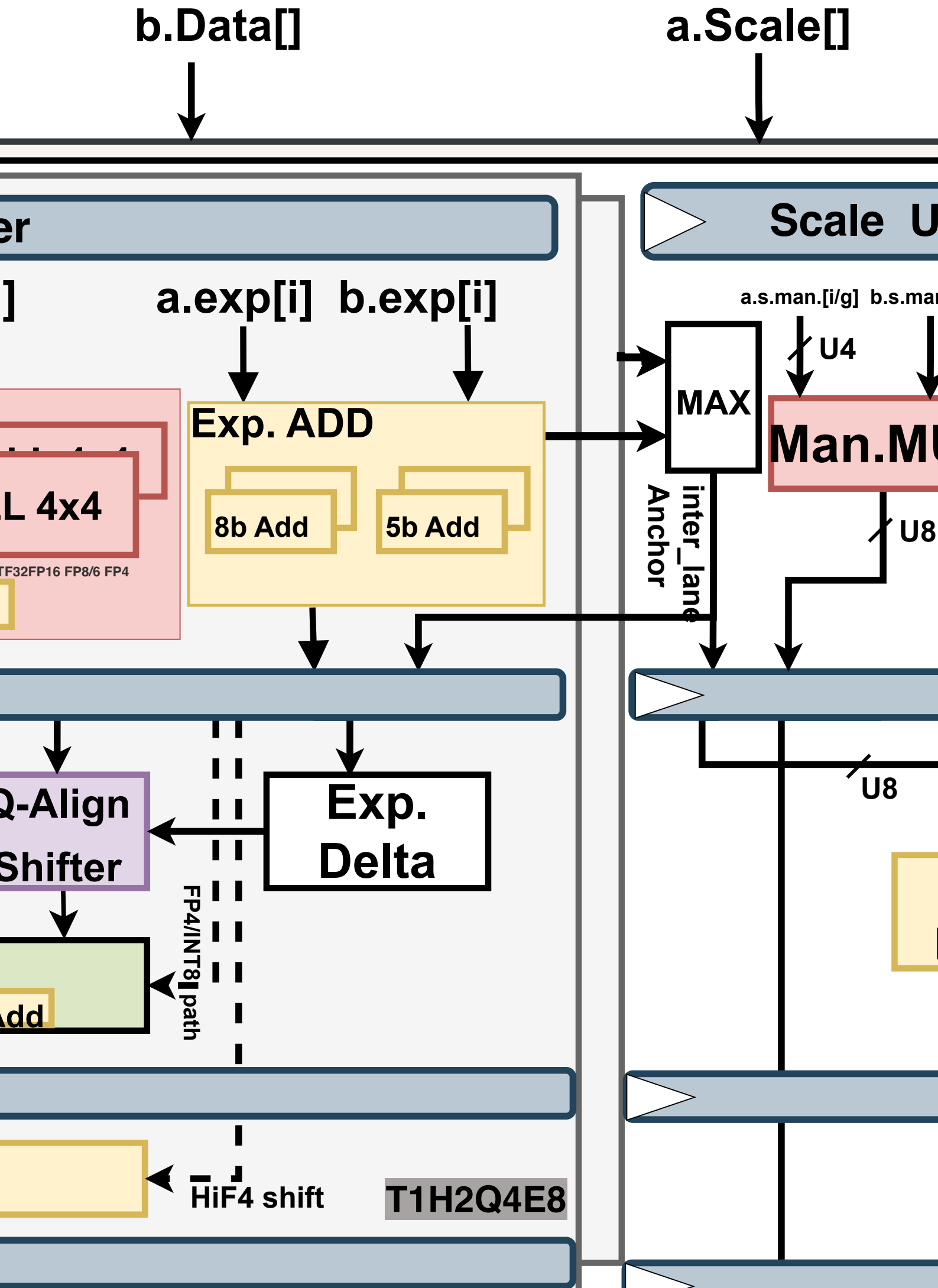
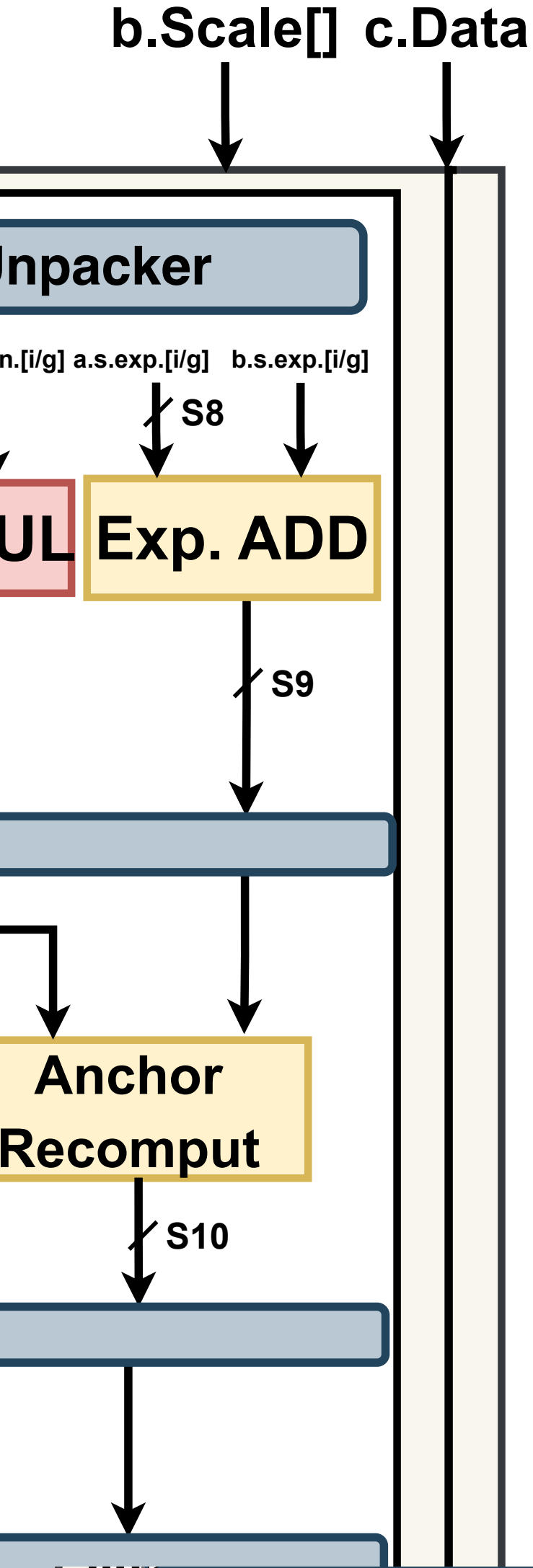
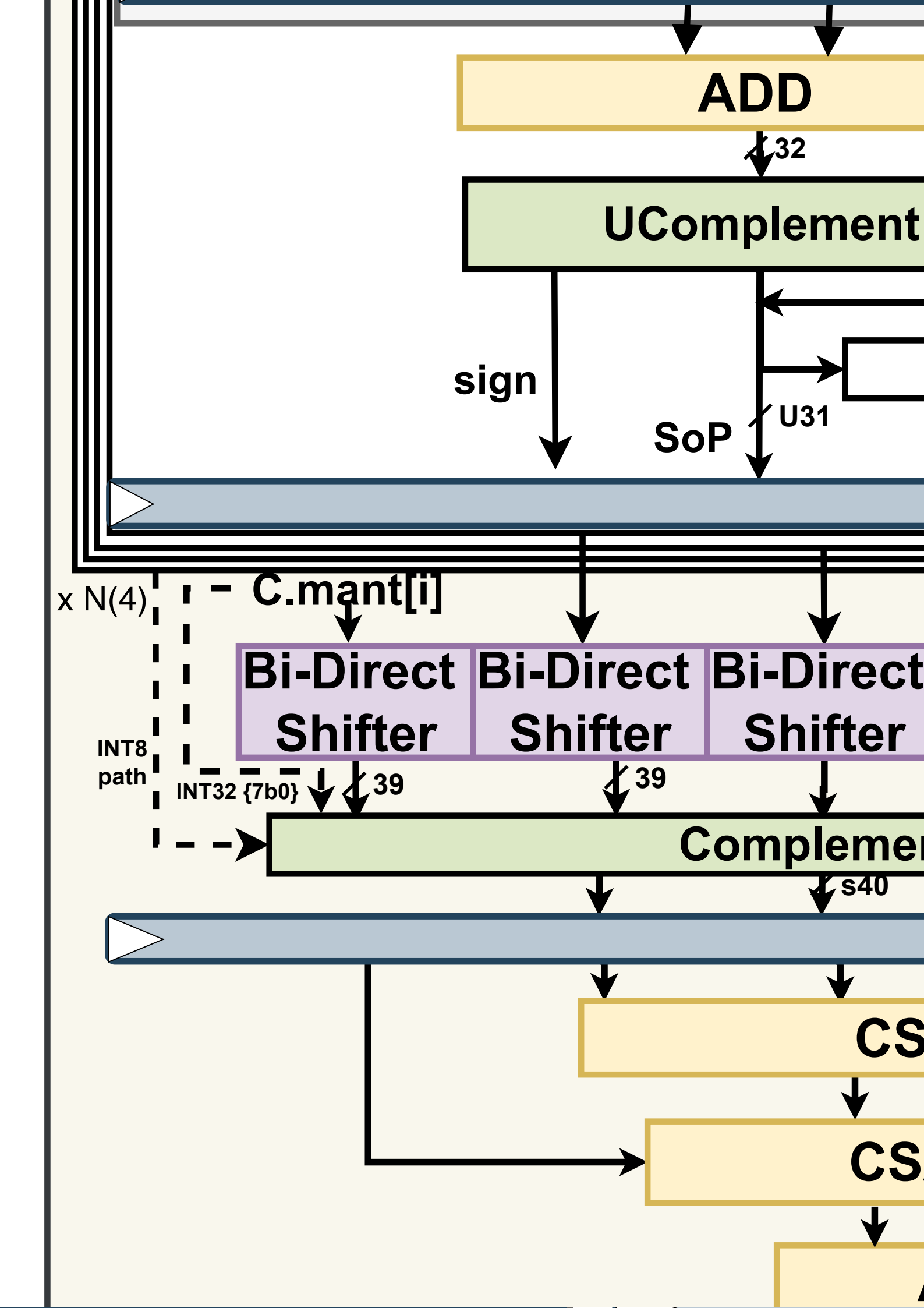
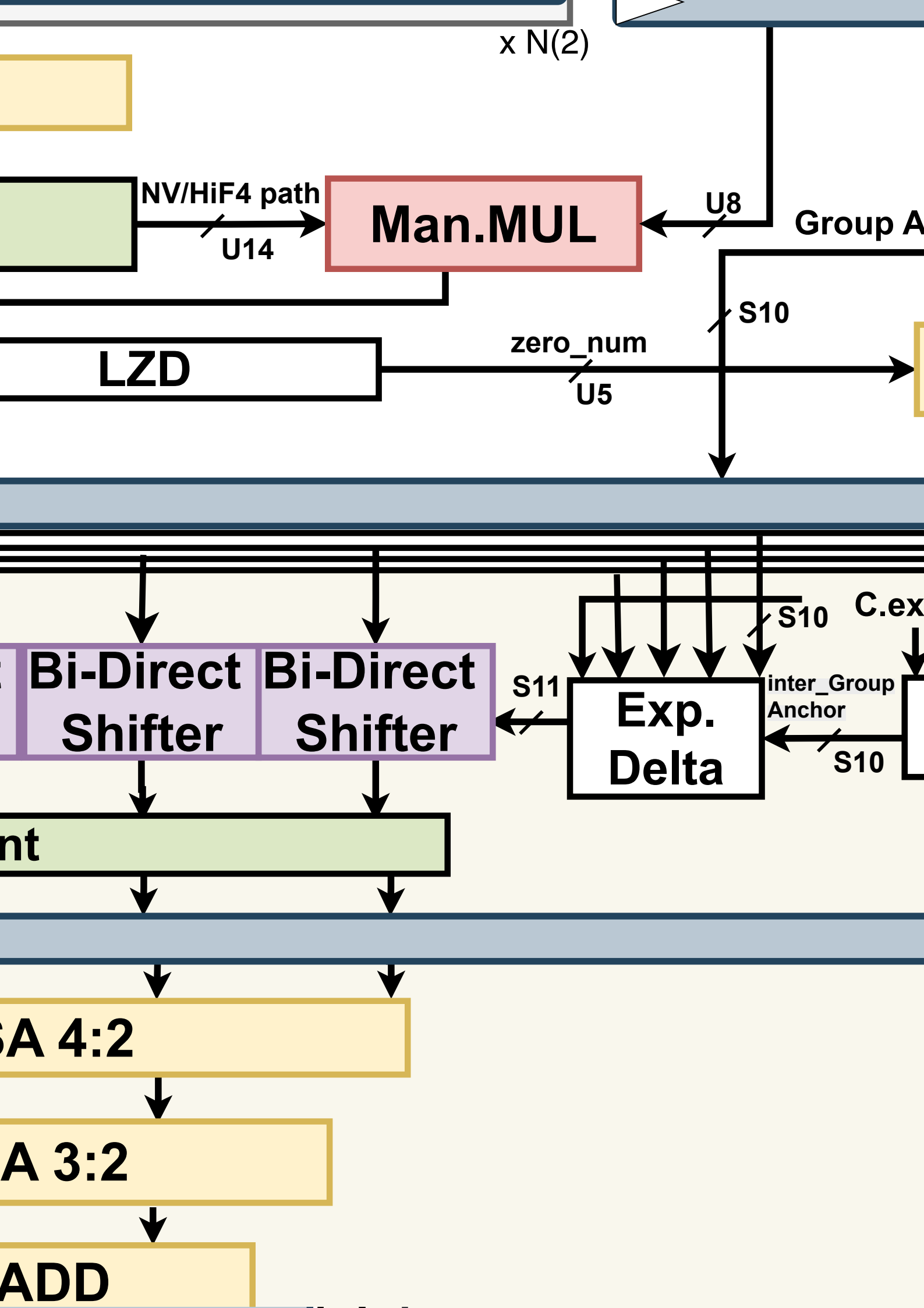
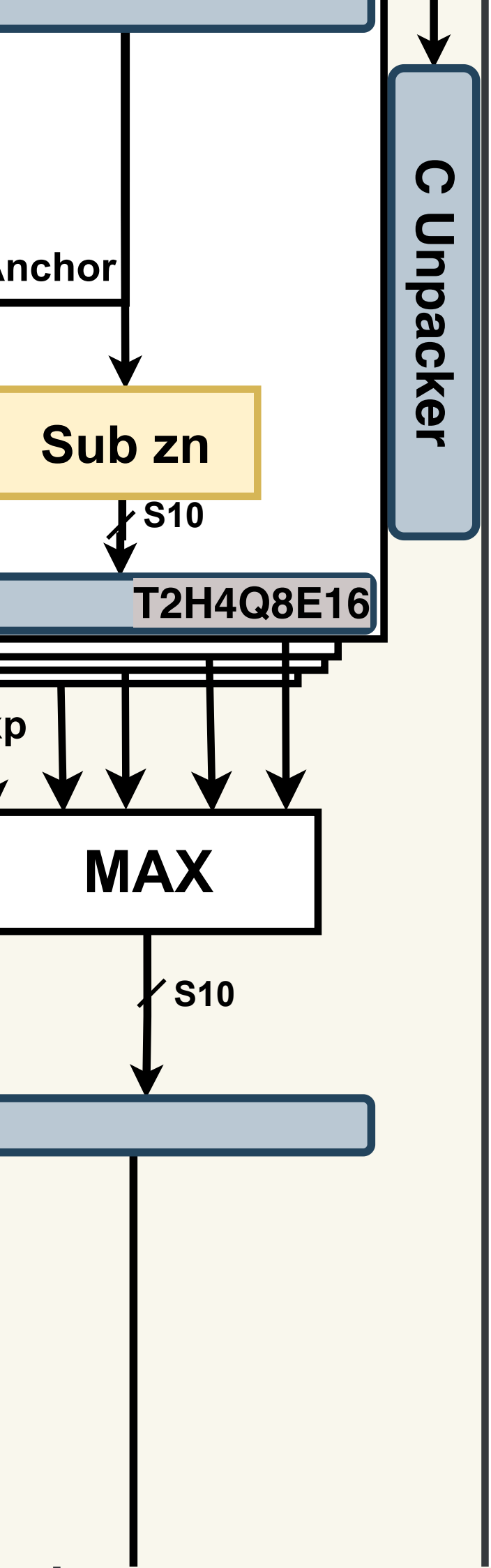

a.Data[]
Data Unpacke
a.sign[i] b.sign[i]
a.mant[i]
b.mant[i
Sign XOR
Man.MUL
HH 8x8
HL 8x4
LH 4x8
TF32FP16BF16INT8 FP4
TF32FP16INT8FP4
TF32FP16INT8FP8/6 FP4
Product Shift Add
THQ-Align Shifter
HQ-Align Shifter
Q-Align Shifter
28
Complement
FP4 Add
FP4 Add
FP4 Add
29
Reduce ADD Tree
31

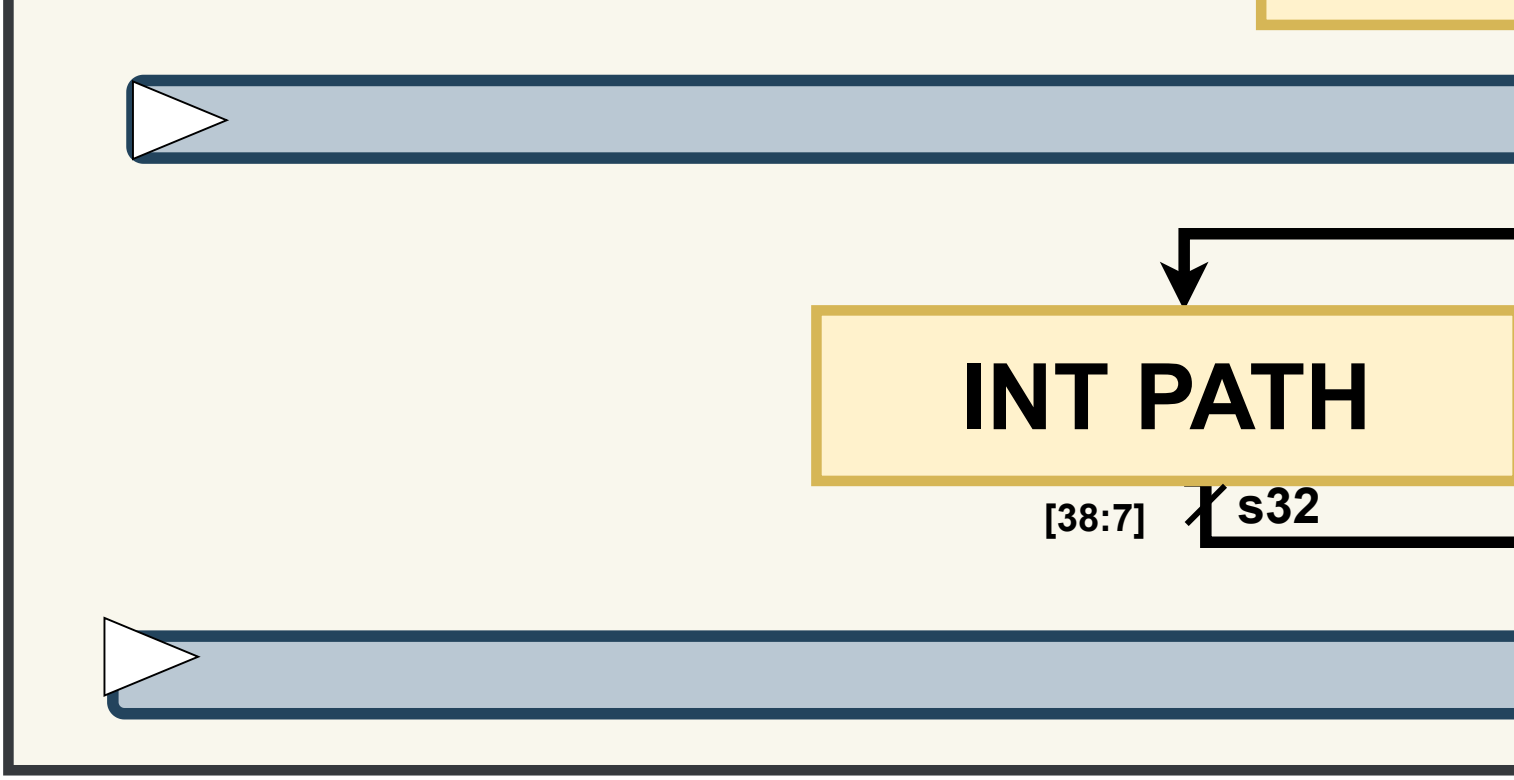

b.Data[]
a.Scale[]
Scale U
a.exp[i] b.exp[i]
a.s.man.[i/g] b.s.man
U4
MAX
Exp. ADD
8b Add
5b Add
inter_lane
Anchor
U8
Exp.
Delta
FP4/INT8 path
HiF4 shift
T1H2Q4E8

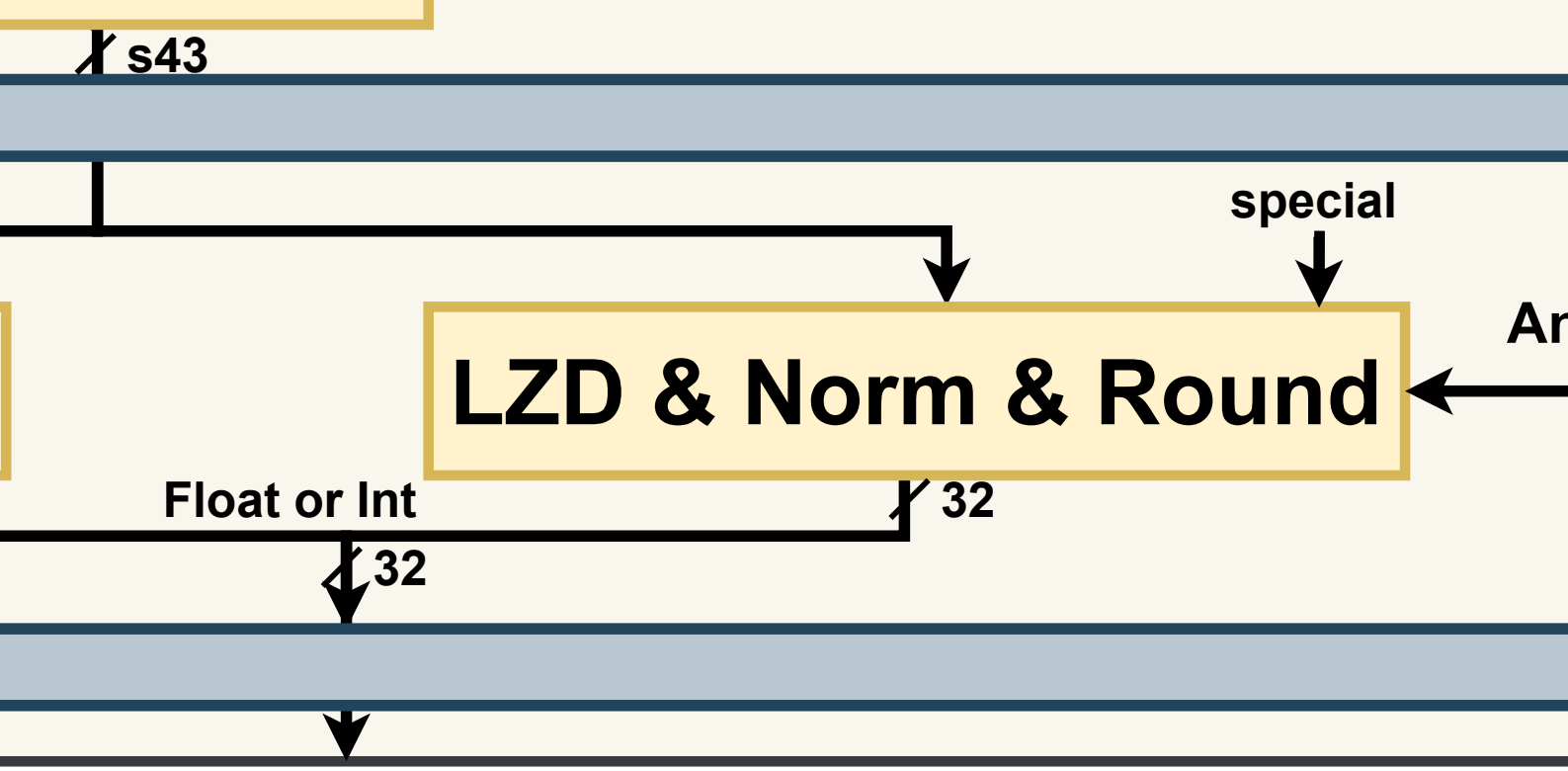

b.Scale[]
c.Data
npacker
n.[i/g]
a.s.exp.[i/g]
b.s.exp.[i/g]
S8
UL
Exp. ADD
S9
Anchor
Recomput
S10

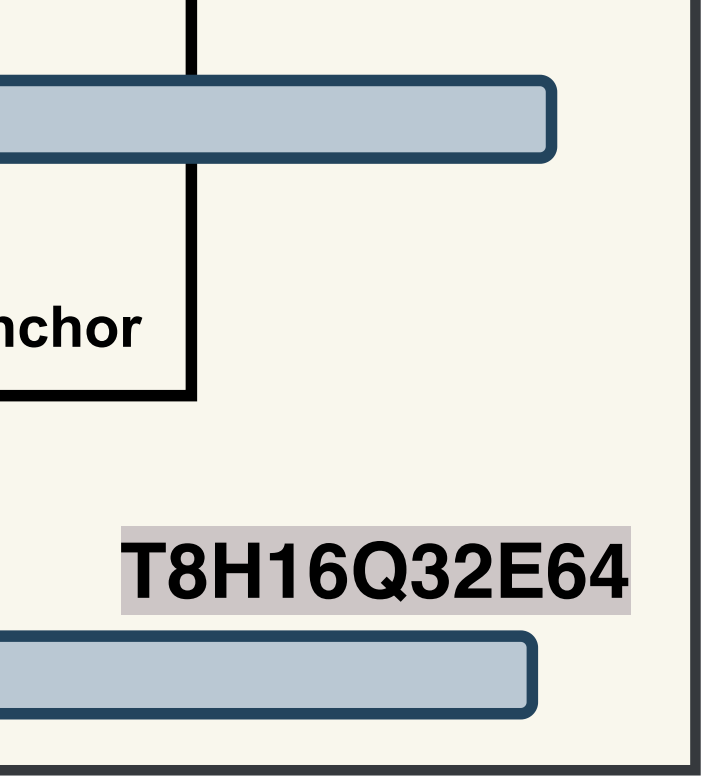

ADD
32
UComplement
sign
SoP
U31
x N(4)
C.mant[i]
Bi-Direct Shifter
Bi-Direct Shifter
Bi-Direct Shifter
INT8 path
INT32 {7b0}
39
39
Compleme
s40
CS
CS


x N(2)
NV/HiF4 path
U14
Man.MUL
U8
Group A
S10
zero_num
U5
LZD
S10
C.ex
Bi-Direct Shifter
Bi-Direct Shifter
S11
Exp. Delta
inter_Group Anchor
S10
nt
A 4:2
A 3:2
ADD


C Unpacker
nchor
Sub zn
S10
T2H4Q8E16
p
MAX
S10


INT PATH
[38:7]
s32


s43
special
LZD & Norm & Round
Ar
Float or Int
32
32


nchor
T8H16Q32E64